

EPISTEMIC WARRANT PROTOCOL

What an agent is justified in accepting — and why

EWP-0.2.0 · Policy reference-v1 · 22 September 2026

MIT · Python 3.12+ · Canonical 14 · Pathological 12 · Goldens 26

EWP defines the deterministic boundary between what an AI agent's memory contains and what the agent is epistemically justified in accepting.

Memory is evidence, not truth.

Stores keep evidence. Warrant is computed. Action is a later gate.

Stores adapt to the protocol. The protocol does not inherit the store's epistemology.

1. What this is

EWP-0.2.0 is an epistemic protocol, not an action-authorization framework. Other projects use warrant to describe permission to act. EWP uses epistemic warrant to describe what an agent is justified in accepting. `may_act()` is deliberately a later gate.

```

Memory / Evidence Store
    |
    v
EvidenceView
    |
    v
warrant_now(view, policy, evaluated_at)
    |
    v
WarrantView
    |
    v
may_act(warrant, action, risk_policy)
    |
    v
MAY_ACT | REQUIRE_CONFIRMATION | DENY

```

Quick start

```

python3 tests/ci.py
python3 tests/report.py

```

```

from protocol.types import Policy
from protocol.warrant import warrant_now
from protocol.fixtures import fixture_verified_current, EVAL
print(warrant_now(fixture_verified_current(), Policy(), EVAL).warrant)

```

There is no packaged install yet. The repository itself is currently the reference implementation and conformance suite.

2. Why this exists

Agents keep conversations, observations, documents, tool results, inferred facts, summaries, and prior decisions. Remembering something is not the same as knowing it is true. Most systems store and retrieve those records. EWP asks: given the evidence available at time T, under policy P, what may this agent accept — and why?

Common collapse	EWP split
stored = true	stored != believed
retrieved = important	retrieved != complete
repeated = corroborated	repeated != independently sourced
summarized = verified	summarized != externally checked
current = warranted	store-local current != warrant
believed = safe to act on	warranted != authorized to act

The distinctions the protocol refuses to collapse.

3. The model

Evidence interchange

Stores expose assertions, evidence, provenance, lineage, conflicts, and verification records as an EvidenceView. Local current-fact, invalidation, ranking, and confidence machinery are inputs. They are not automatically agent beliefs.

Warrant evaluation

warrant_now(evidence_view, policy, evaluated_at) is deterministic. No I/O. No LLM. Same EvidenceView + same policy version + same evaluated_at yields the same WarrantView. Warrant is a function, not a stored truth field.

Decision gating

An agent may be justified in accepting that a customer requested cancellation and still need confirmation before cancelling a \$2M contract. Risk, reversibility, authorization, privacy, money, and safety belong in action policy.

4. WarrantView axes

EWP does not collapse state into TRUE, VERIFIED, or DISPUTED. A proposition can be externally checked and still disputed; accepted and stale; well evidenced with an open conflict.

Axis	Values
acceptance	UNACCEPTED / TENTATIVE / ACCEPTED
conflict	NONE / OPEN / RESOLVED
verification	NONE / INDIRECT / EXTERNAL / HUMAN
currency	CURRENT / STALE / SUPERSEDED
sufficiency	SUFFICIENT / INSUFFICIENT / DEGRADED

Verification is a record (method, source, scope, time, result, freshness policy), not a badge. result=opposes opens conflict and blocks ACCEPTED. result=inconclusive cannot raise EXTERNAL or HUMAN. Endogenous origin caps human and external methods at INDIRECT. Scope identity is declared subjects[] only; the scope string is an inspection surface and is not scraped for identifiers. Currency is computed at evaluation time from checks that confer the chosen class. A later summary cannot refresh an old tool observation. History is not rewritten.

5. Lineage and the three confidences

Ten transformations of one source are not ten independent confirmations. The evaluator counts unique lineage_id values on assertions, evidence, and checks — not repetitions.

Kind	Meaning
Assertion confidence	How strongly the source or extractor stated the claim
Retrieval score	How relevant a record looks to this query
Warrant strength	How strongly policy permits acceptance

Implementations MUST NOT use retrieval relevance, repetition count, memory strength, or assertion confidence as substitutes for warrant.

6. Core invariants

- Assertions are not beliefs.
- Beliefs are not truth.
- Warrant is computed, not persisted as truth.
- Verification is evidence with method, scope, source, and time.
- Endogenous processing cannot manufacture external verification.
- Derivation cannot manufacture provenance.
- Contradiction must survive storage and retrieval.
- Incomplete retrieval must be visible as sufficiency=DEGRADED.
- Epistemic policy and action policy are separate.
- Semantically equivalent evidence must yield equivalent warrant independent of storage substrate.

Retrieval, summarization, dreaming, consolidation, reranking, repetition, graph propagation, LLM critique, and multi-agent agreement may reorganize. They may not turn verification=NONE into EXTERNAL without new external evidence. If search drops a contradictor, sufficiency is DEGRADED.

7. Platforms

OpenClaw

```
openclaw mcp add ewp --url http://127.0.0.1:8765/mcp
```

OpenClaw object	Role under EWP
Daily notes, transcripts	Raw lineage. Append. Do not rewrite.
USER.md	Working checkout of preferences. Regenerable.
MEMORY.md	Persona briefing from WarrantViews. Never authoritative.
Dreaming / promotion	Candidate generator. Must not raise verification class.
Local SQLite state	Cache. Not a second ledger.

Dreaming may rewrite MEMORY.md. EWP treats that rewrite as a new assertion, not as verification. Compression must not turn "X is disputed" into "X."

Claude, Codex, other MCP clients

Same contract. `python3 -m protocol.mcp_server --stdio --allow-ingest` speaks Content-Length framed JSON-RPC (MCP). `--http 127.0.0.1:8765` serves plain JSON-RPC POST /mcp for OpenClaw. HTTP is not framed MCP. Trusted writes require a server-side ingest role (`--allow-ingest` on stdio, or Authorization: Bearer matching `--ingest-token` on HTTP) plus `ingest_attestation=true`. `ewp_may_act` refuses a caller-built WarrantView. The pre-freeze claim/confidence sketch stays in `historical/docs/MCP_CONTRACT.md` and is not the shipped façade.

```
python3 -m protocol.mcp_server --stdio --allow-ingest --db ./ewp.sqlite
python3 -m protocol.mcp_server --http 127.0.0.1:8765 --ingest-token "$EWP_INGEST_TOKEN" --db ./ewp.sqlite
openclaw mcp add ewp --url http://127.0.0.1:8765/mcp
```

Graphiti, Mem0, Particles, SQLite

Graphiti can be used as a temporal/entity evidence substrate or mirror. Inject `lineage_id` in episode metadata. `invalid_at` is store-local. `valid_at` is not a verification check. Search collapse marks the view DEGRADED. The frozen suite uses fake Graphiti-shaped records. Live Graphiti and Mem0 mappings live in `protocol/graphiti_client_adapter.py` and `protocol/mem0_adapter.py`; notes in `docs/implementer/LIVE_ADAPTERS.md`. Live graphiti-core 0.30.2 is not validated. Mem0 default origin is extract; retrieval score is not warrant. Particles is a good immutable substrate — agents write only through EWP. SQLite and JSON prove store neutrality.

Graphiti adapts to the protocol. The protocol does not adapt to Graphiti.

8. Why EWP sits above the store

System	Primary concern	EWP adds
Graphiti / Zep	Temporal knowledge graph and retrieval	Store-independent warrant evaluation
Mem0	Agent memory storage and retrieval	Evidence lineage and deterministic warrant policy
OpenClaw native memory	Inspectable agent context and memory	Separation of persona, evidence, and warrant
EWP	Epistemic evaluation	Not a general-purpose memory store

9. Conformance

```
python3 tests/ci.py
python3 tests/report.py
python3 tests/runner.py
python3 tests/runner_pathological.py
```

CI enforces fixture, evaluator, and golden lock hashes, all 26 goldens, SQLite = JSON, fake-Graphiti isolation, and the laundering pack. Changing a golden or the evaluator requires a protocol or policy bump, then `python3 tests/ci.py --write-lock`.

Class	Meaning
INGEST_LOSS	Store dropped assertions, sources, lineage, checks, or conflicts.
ADAPTER_MAP_LOSS	Store has the rows; EvidenceView is incomplete.
WARRANT_MISMATCH	Reconstructed view yields a WarrantView that is not the golden.
RETRIEVAL_LOSS	Search omitted evidence and did not mark DEGRADED.
EXPECTED_DIVERGENCE	Store current differs from warrant_now. Observation, not failure.

Pathological pack: false supersession, open contradiction, real temporal upgrade, same text / two lineages, three wordings / one lineage, store-current vs verified, weak invalidating strong, maintenance expiry that must not erase a check, retrieval-induced false consensus, extractor polarity flip on one lineage, canonical-without-verification, invalidation cycles.

10. Freeze

```
Epistemic Warrant Protocol EWP-0.2.0
Policy: reference-v1
Canonical: 14/14      Pathological: 12/12
SQLite PASS    JSON PASS    Fake Graphiti PASS
graphiti-core 0.30.2 - NOT VALIDATED
Fixture set sha256:
910b6e98bee3148460f15303810c8e4c447721252b02c7f4805f3c0ce75b98db
Evaluator set sha256:
b1067194eb02a2cc419f8ee56e92b380eb1840f36e3d25a2d1f311989b9ff77a
Golden set sha256:
95f26b124ac813ef7b6f895bd43c20832f2026bfb8a25513ce6bfd7302088dd3
```

A store that produces a different answer has an adapter or conformance problem, not a license to move the goldens.

11. What EWP is not

Not a vector database, knowledge graph, memory engine, truth oracle, LLM fact-checker, or authorization framework. It does not ask what is ultimately true. It asks a narrower, computable question: given this bounded evidence view, at this time, under this versioned policy, what may the agent accept — and why?

v0.2.0 tightened generic scope binding, refused future-dated checks at T, persisted SQLite view completeness, and narrowed the serialized WarrantView. Later 0.2.x hardening shipped the MCP façade with framed stdio and an ingest role, time-filtered assertions and evidence, and required subjects[] for scope caps. The 26 goldens stay. New stores may reveal adapter bugs; they do not redefine warrant.

Stores keep evidence. Warrant is computed. Action is a later gate.